

An Introduction to Measure Concentration Phenomenon

Po Hsuan Chen

2026-09-20

1.1 The phenomenon

Measure concentration (P. Lévy, V. Milman): in many high-dimensional spaces,

- a set containing half of the space, made a little bigger, contains almost all of the space;
- equivalently, a function that cannot change fast is almost constant.

Interactive introduction: cclion61-droid.github.io/mc-interactive-slides

To make this precise we need

1. a notion of **distance**: metric spaces;
2. a notion of **size**: measures;
3. a number that measures concentration: the **concentration function**.

2.1 Metric spaces

Definition 2.1: A **metric** on a set X is a function $d : X \times X \rightarrow [0, \infty)$ such that for all $x, y, z \in X$:

1. $d(x, y) = 0 \iff x = y$;
2. $d(x, y) = d(y, x)$;
3. $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality).

The pair (X, d) is a **metric space**.

Example ($\mathbb{R}^1$). $d(x, y) = |x - y|$. The triangle inequality is $|x - z| \leq |x - y| + |y - z|$, and the open ball is $B(x, r) = (x - r, x + r)$.

2.2 Example: the sphere

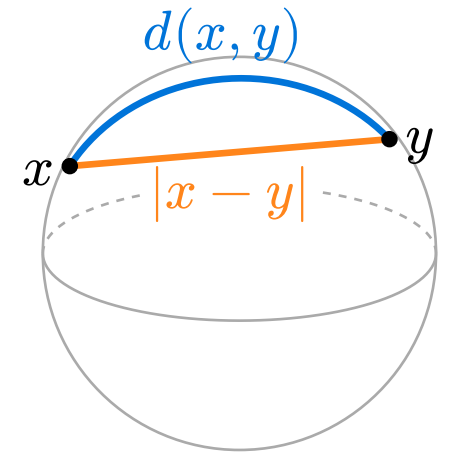
On $S^n \subset \mathbb{R}^{n+1}$ there are two natural metrics:

- the **straight-line distance** $|x - y|$, inherited from $\mathbb{R}^{n+1}$;
- the **geodesic** metric $d(x, y) = \arccos \langle x, y \rangle$, the length of the shortest arc of a great circle from x to y .

They are related by $|x - y| = 2 \sin(d(x, y)/2)$, so

$$\frac{2}{\pi} d(x, y) \leq |x - y| \leq d(x, y).$$

From now on S^n carries the geodesic metric d .

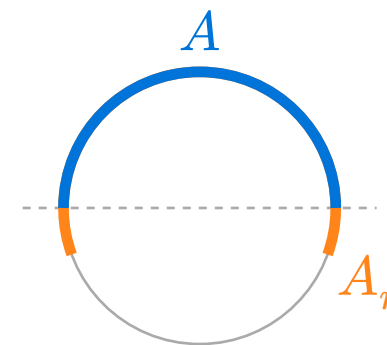


2.3 r -neighbourhoods

Definition 2.2: For $A \subseteq X$ and $r > 0$, let $d(x, A) := \inf_{a \in A} d(x, a)$ and

$$A_r := \{x \in X : d(x, A) < r\}.$$

- In $\mathbb{R}$: $A = [a, b]$ gives $A_r = (a - r, b + r)$.
- On S^n : the northern hemisphere $A = \{x_{n+1} \geq 0\}$ gives $A_r = \{x_{n+1} > -\sin r\}$: everything except a cap of radius $\pi/2 - r$ around the south pole.



3.1 σ -algebras and measures

Definition 3.1: A **σ -algebra** on X is a collection $\mathcal{F}$ of subsets of X with $\emptyset \in \mathcal{F}$, closed under complements and countable unions.

Definition 3.2: A **measure** is a map $\mu : \mathcal{F} \rightarrow [0, \infty]$ with $\mu(\emptyset) = 0$ and

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i) \quad \text{for pairwise disjoint } A_i.$$

It is a **probability measure** if $\mu(X) = 1$.

On a metric space we use the **Borel** σ -algebra: the smallest one containing all open sets. (Not every subset of $\mathbb{R}$ can be given a length.)

3.2 Example: $\mathbb{R}^1$

The **Lebesgue measure** λ is the length: $\lambda([a, b]) = b - a$.

- $\lambda(\{x\}) = 0$, hence $\lambda(\mathbb{Q}) = \sum_{q \in \mathbb{Q}} \lambda(\{q\}) = 0$ by countable additivity.
- $\lambda(\mathbb{R}) = \infty$, so λ is not a probability measure.

A probability measure out of it: the **uniform** measure on $[0, 1]$, $\mu(A) = \lambda(A \cap [0, 1])$.

3.3 Example: the sphere

The **normalised surface measure** σ on S^n : $\sigma(A) = \text{area}(A) / \text{area}(S^n)$.

- It is invariant under rotations: $\sigma(UA) = \sigma(A)$ for every orthogonal U .
- With $\theta \in [0, \pi]$ the angle from the north pole,

$$\sigma(\{a \leq \theta \leq b\}) = \frac{\int_a^b \sin^{n-1} \theta \, d\theta}{\int_0^\pi \sin^{n-1} \theta \, d\theta}.$$

- For $n = 2$ this is $(\cos a - \cos b)/2$: the area of a band is proportional to its height (Archimedes).

3.4 Metric measure spaces

Definition 3.3: A **metric measure space** (mm-space) (X, d, μ) is a metric space (X, d) with a Borel probability measure μ .

Examples:

- $([0, 1], |\cdot|, \lambda)$;
- (S^n, d, σ) with the geodesic metric and normalised area.

4.1 Half of an interval and its r -neighbourhood

Take $X = [0, 1]$ with λ and $A = [0, 1/2]$, so $\lambda(A) = 1/2$. Then $A_r = [0, 1/2 + r)$ and

$$1 - \lambda(A_r) = \frac{1}{2} - r \quad (r \leq 1/2).$$

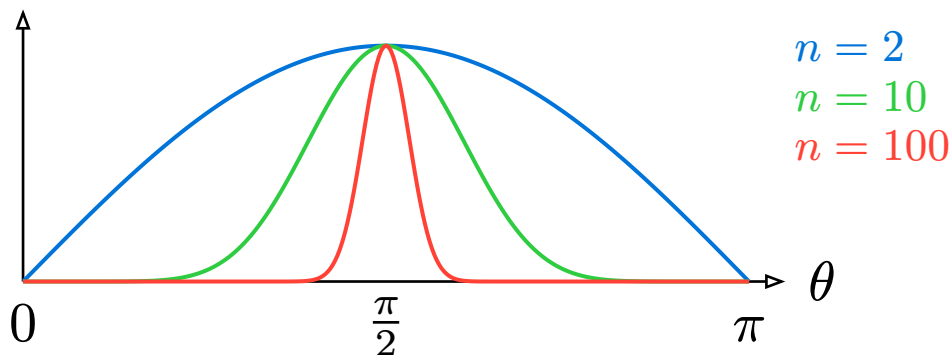
On the circle S^1 , a half circle A gives $1 - \sigma(A_r) = \frac{1}{2} - \frac{r}{\pi}$.

The mass left outside A_r decreases only **linearly** in r : no concentration.

4.2 Half of a sphere and its r -neighbourhood

$A = \{x_{n+1} \geq 0\}$ has $\sigma(A) = 1/2$, and $S^n \setminus A_r$ is the cap $\theta \geq \pi/2 + r$:

$$1 - \sigma(A_r) = \frac{\int_{\pi/2+r}^{\pi} \sin^{n-1} \theta \, d\theta}{\int_0^{\pi} \sin^{n-1} \theta \, d\theta}.$$



Near $\theta = \pi/2$,

$$\sin^{n-1} \theta \approx e^{-(n-1)(\theta-\pi/2)^2/2},$$

so almost all mass lies within $\approx 1/\sqrt{n}$ of the equator.

4.3 Lévy's inequality

Theorem 4.1 (Lévy): If $A \subseteq S^n$ is Borel and C is a cap with $\sigma(C) = \sigma(A)$, then $\sigma(A_r) \geq \sigma(C_r)$ for all $r > 0$.

Corollary 4.2: If $\sigma(A) \geq 1/2$, then $1 - \sigma(A_r) \leq e^{-(n-1)r^2/2}$ for all $r > 0$.

On S^{1000} with $r = 0.1$: $1 - \sigma(A_r) \leq e^{-4.995} < 0.007$, **whatever** the half A is. Compare $[0, 1]$, where $1 - \lambda(A_r) = 0.4$.

5.1 Definition

Definition 5.1: The **concentration function** of an mm-space (X, d, μ) is

$$\alpha_\mu(r) := \sup \left\{ 1 - \mu(A_r) : A \subseteq X \text{ Borel}, \mu(A) \geq \frac{1}{2} \right\}, \quad r > 0.$$

α_μ is non-increasing and $\alpha_\mu \leq 1/2$; small $\alpha_\mu(r)$ means strong concentration.

- $([0, 1], \lambda)$: $\alpha(r) = \max(1/2 - r, 0)$;
- (S^1, σ) : $\alpha(r) = \max(1/2 - r/\pi, 0)$;
- (S^n, σ) : $\alpha(r) \leq e^{-(n-1)r^2/2}$ (Lévy).

5.2 Lévy families

Definition 5.2: A sequence of mm-spaces (X_n, d_n, μ_n) is a **Lévy family** if

$$\alpha_{\mu_n}(r) \rightarrow 0 \quad \text{for every } r > 0.$$

- (S^n, d, σ) , $n = 1, 2, \dots$ is a Lévy family, since $e^{-(n-1)r^2/2} \rightarrow 0$.
- The constant sequence $([0, 1], |\cdot|, \lambda)$ is not: $\alpha(r) = 1/2 - r$ for all n .

5.3 Example: a hypersurface of revolution

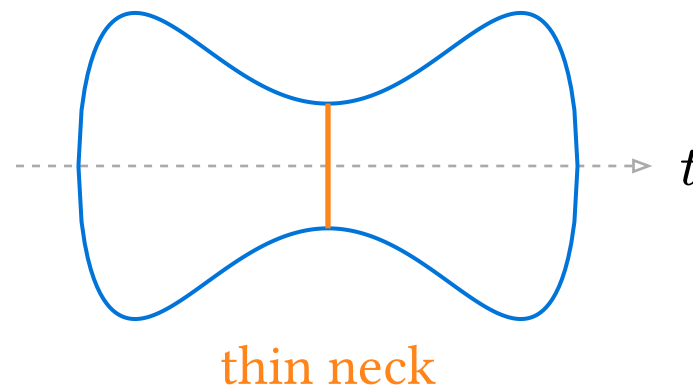
Rotating a profile $f : [a, b] \rightarrow (0, \infty)$ about an axis gives

$$\Sigma_f := \{(t, f(t)u) : t \in [a, b], u \in S^{n-1}\} \subset \mathbb{R}^{n+1},$$

with the geodesic distance of Σ_f and the normalised area

$$\mu_f \propto f(t)^{n-1} \sqrt{1 + f'(t)^2} dt du.$$

- $f(t) = \sqrt{1 - t^2}$ gives back S^n : the density is $(1 - t^2)^{(n-2)/2}$, that is, $\sin^{n-1} \theta d\theta$.
- A thin neck splits the mass into two halves: making one of them a little bigger gains almost nothing, so $\alpha_{\mu_f}(r) \approx 1/2$ for small r : no concentration.



6.1 The Heisenberg group

On $\mathbb{H}^n = \mathbb{R}^{2n+1}$ with coordinates (x, y, t) , $x, y \in \mathbb{R}^n$:

$$X_i := \partial_{x_i} - \frac{y_i}{2} \partial_t, \quad Y_i := \partial_{y_i} + \frac{x_i}{2} \partial_t, \quad T := \partial_t,$$

$$\Theta := dt + \frac{1}{2} \sum_{i=1}^n (y_i dx^i - x_i dy^i), \quad \xi := \ker \Theta = \text{span}\{X_i, Y_i\},$$

with $[X_i, Y_j] = \delta_{ij}T$ and g the sub-Riemannian metric on ξ making $\{X_i, Y_i\}$ orthonormal.

6.2 The sphere in the Heisenberg group

The Euclidean unit sphere $\Sigma := S^{2n} = \{|x|^2 + |y|^2 + t^2 = 1\}$ inherits

- a distribution $\mathcal{H} := \xi \cap T\Sigma$ with the metric $g|_{\mathcal{H}}$, hence a Carnot–Carathéodory distance d_{cc} (horizontal curves only);
- the p-area measure, normalised: $\mu \propto \sqrt{(1 - t^2)(1 + t^2/4)} \, d\sigma_0$.

Question. What is the concentration function of $\Sigma := (S^{2n}, d_{\text{cc}}, \mu)$?

6.3 From sub-Riemannian structure to Riemannian structure

The sub-Riemannian structure on Σ is too difficult to study directly: it is not a Riemannian manifold, so the usual estimates cannot be applied to it.

But it is a limit of Riemannian ones: for $L > 0$,

$$g_L := \sum_i \left((dx^i)^2 + (dy^i)^2 \right) + L\Theta^2, \quad |v|_{g_L}^2 = |v_\xi|_g^2 + L\Theta(v)^2,$$

where v_ξ is the horizontal part of v . Horizontal vectors have L -free length; everything transverse to ξ is inflated.

On Σ this gives compact Riemannian mm-spaces $\Sigma_L := (S^{2n}, d_L^\Sigma, \mu_L)$, with d_L^Σ the intrinsic distance of $(\Sigma, g_L|_\Sigma)$ and μ_L its normalised area. As $L \rightarrow \infty$:

$$d_L^\Sigma \uparrow d_{cc}, \quad \mu_L \rightarrow \mu.$$

6.4 The family

A computation in coordinates $(u, t) \in S^{2n-1} \times (-1, 1)$, $(x, y) = \sqrt{1-t^2}u$, with $\eta := \left(\sum_i (x_i dy^i - y_i dx^i)\right)|_{S^{2n-1}}$ gives

$$g_L|_{\Sigma} = \frac{t^2}{1-t^2} dt^2 + (1-t^2)g_{S^{2n-1}} + L\left(dt - \frac{1}{2}(1-t^2)\eta\right)^2,$$

$$\text{Vol}_{\Sigma, g_L} = W d\sigma_0, \quad W := \sqrt{L(1-t^2)(1+t^2/4) + t^2}.$$

Normalising gives $\mu_L \propto W d\sigma_0$, and $L^{-1/2}W \rightarrow \sqrt{(1-t^2)(1+t^2/4)}$ gives $\mu_L \rightarrow \mu$.
Hence

$$\Sigma_L = (S^{2n}, d_L^{\Sigma}, \mu_L) \longrightarrow \Sigma = (S^{2n}, d_{\text{cc}}, \mu).$$

6.5 Measured Gromov–Hausdorff convergence

Definition 6.1: $f : X \rightarrow Y$ is ε -isometric if

$$|d_Y(f(x), f(y)) - d_X(x, y)| \leq \varepsilon \quad \text{for all } x, y.$$

Compact mm-spaces (X_n, d_n, μ_n) converge to (X, d, μ) in the *measured Gromov–Hausdorff* sense if there are Borel ε_n -isometric maps $f_n : X_n \rightarrow X$, $\varepsilon_n \rightarrow 0$, with

$$\nu_n := (f_n)_* \mu_n \rightarrow \mu \quad \text{weakly.}$$

Here $(f_n)_* \mu_n(B) := \mu_n(f_n^{-1}(B))$, and $\nu_n \rightarrow \mu$ weakly means $\int \varphi d\nu_n \rightarrow \int \varphi d\mu$ for every continuous $\varphi : X \rightarrow \mathbb{R}$.

6.6 Question

$\Sigma_L \rightarrow \Sigma$ is exactly measured Gromov–Hausdorff convergence. So:

Is the concentration function α continuous along mGH convergence?

Result. For $\alpha := \alpha_\mu$, $\alpha_n := \alpha_{\mu_n}$ and $\alpha(r^\pm) := \lim_{s \rightarrow r^\pm} \alpha(s)$:

$$\alpha(r^+) \leq \liminf_{n \rightarrow \infty} \alpha_n(r) \leq \limsup_{n \rightarrow \infty} \alpha_n(r) \leq \alpha(r^-), \quad r > 0.$$

The last inequality is unconditional; the first needs one hypothesis.

6.7 What weak convergence gives — and what it does not

Theorem 6.2 (Portmanteau):

$$\begin{aligned}\nu_n \rightarrow \mu \text{ weakly} &\iff \liminf_n \nu_n(U) \geq \mu(U) \text{ for open } U \\ &\iff \limsup_n \nu_n(F) \leq \mu(F) \text{ for closed } F.\end{aligned}$$

Not obvious: $\alpha_\mu(r) = \sup\{1 - \mu(A_r) : \mu(A) \geq \frac{1}{2}\}$ is a supremum over all Borel sets, the threshold is pinned at $\frac{1}{2}$, and weak convergence controls open sets in one direction only.

Theorem 6.3 (Ledoux): If $\mu(A) \geq \theta > 0$ and $\alpha_\mu(r_0) < \theta$, then $1 - \mu(A_{r_0+r}) \leq \alpha_\mu(r)$ for every $r > 0$.

A threshold $\theta < \frac{1}{2}$ is converted back to $\frac{1}{2}$, at the cost of a shift r_0 .

6.8 The upper bound: unconditional

Lemma 6.4 ([Chen, Group 5]): For every $r > 0$ and every $u \in (0, r)$,

$$\limsup_{n \rightarrow \infty} \alpha_n(r) \leq \alpha(u), \quad \text{hence} \quad \limsup_{n \rightarrow \infty} \alpha_n(r) \leq \alpha(r^-).$$

No hypothesis on the above inequality.

Proof: Almost optimal sets $A^{(n)} \subseteq X_n$ have images $f_n(A^{(n)})$ that need not be Borel, and need not converge. Replace them by the closed sets $C_n := \overline{f_n(A^{(n)})}$ and use Blaschke's selection theorem: $(\mathcal{F}(X), d_H)$ is compact, so $C_n \rightarrow C$ in Hausdorff distance along a subsequence. Portmanteau on the closed $\overline{C_\eta}$ gives $\mu(C) \geq \frac{1}{2}$; Portmanteau on the open C_u finishes. ■

Compactness of the *limit* space is used here, and again in the gap proposition below.

6.9 The lower bound

Lemma 6.5 ([Chen, Group 5]): Fix $r > 0$, $0 < \varepsilon < \frac{1}{2}$; let $A \subseteq X$ be open with $\mu(A) \geq \frac{1}{2}$ and $B := f_n^{-1}(A)$. If

- (a) $d > 1$ with $r/(r - \varepsilon_n) < d$;
- (b) $\nu_n(A) \geq \mu(A) - \varepsilon$ and $\nu_n(\overline{A_r}) \leq \mu(\overline{A_r}) + \varepsilon$ (true for large n)
- (c) $r_n > 0$ with $\alpha_n(r_n) < \frac{1}{2} - \varepsilon$ and $r/d - r_n > 0$,

then

$$1 - \mu(\overline{A_r}) \leq \alpha_n(r/d - r_n) + \varepsilon.$$

6.10 Reading the lower bound

(a) absorbs the isometry defect; (c) is Ledoux's lemma applied *inside* X_n , with threshold $\frac{1}{2} - \varepsilon$; the mass defect ε enters additively. Nothing is asymptotic yet.

Corollary 6.6 ([Chen, Group 5]): For $r > 0$, $0 < \varepsilon < \frac{1}{2}$ there is N with $\alpha(r^+) \leq \alpha_n(r/d - r_n) + 2\varepsilon$ for all $n \geq N$ and all admissible d, r_n .

6.11 The hypothesis: no asymptotic flatness at level 1/2

Everything hinges on whether the Ledoux radius r_n can be taken small. Put

$$\rho_n(\varepsilon) := \inf \left\{ s > 0 : \alpha_n(s) < \frac{1}{2} - \varepsilon \right\}, \quad \rho(\varepsilon) := \limsup_n \rho_n(\varepsilon).$$

Hypothesis 6.7 (NF): $\lim_{\varepsilon \downarrow 0} \rho(\varepsilon) = 0$; equivalently, $\limsup_{n \rightarrow \infty} \alpha_n(s) < \frac{1}{2}$ for every $s > 0$.

A criterion on the limit space alone: $\alpha(s) < \frac{1}{2}$ for all $s > 0 \implies$ (NF).

Proposition 6.8: If $\alpha_\mu(s) = \frac{1}{2}$ for some $s > 0$, then $\text{supp } \mu \subseteq F \cup G$ with F, G disjoint, non-empty, closed, $d(F, G) \geq s/2$ and $\mu(F) = \mu(G) = \frac{1}{2}$. Hence $\text{supp } \mu$ connected $\implies$ (NF).

6.12 The convergence theorem

Theorem 6.9 ([Chen, Group 5]): Assume (NF). Then for every $r > 0$

$$\alpha(r^+) \leq \liminf_{n \rightarrow \infty} \alpha_n(r) \leq \limsup_{n \rightarrow \infty} \alpha_n(r) \leq \alpha(r^-).$$

Consequently $\alpha_n(r) \rightarrow \alpha(r)$ at every continuity point r of α , hence for all but countably many $r > 0$.

Proof: Right inequality: the upper bound. Left: at radius $v > u$ choose $d_n \downarrow 1$ (possible since $\varepsilon_n \rightarrow 0$) and r_n near $\rho(\varepsilon)$ (small by (NF)), so that $v/d_n - r_n > u$; then the corollary to the lower bound and monotonicity give $\alpha(v^+) \leq \alpha_n(u) + 2\varepsilon$. Let $\varepsilon \downarrow 0$, then $v \downarrow u$. ■

6.13 The conclusion

Example (The sub-Riemannian sphere): Let $n \geq 2$ and $L \rightarrow \infty$. Then $\Sigma_L \rightarrow \Sigma$ in the measured Gromov–Hausdorff sense, with $\varepsilon_L = \varepsilon_L^\Sigma \downarrow 0$. Since $\text{supp } \mu = \Sigma$ is connected, the gap proposition gives (NF) and the convergence theorem gives

$$\alpha_{\Sigma_L}(r) \longrightarrow \alpha_\Sigma(r) \quad (L \rightarrow \infty)$$

at every continuity point of α_Σ , hence for all but countably many $r > 0$.